

Practical 1

Aim: Linux System Administration and File Management for Marketing

Problem definition:

You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers where project data, reports, scripts, and user accounts must be managed efficiently. Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team.

Tasks:

1. Identify the current user.
2. Find the current working directory.
3. Display system information.
4. Check system uptime and date.
5. Create a directory named MarketingAnalytics.
6. Create subdirectories: RawData, ProcessedData, Reports.
7. Create data and report files.
8. Add sample content.
9. Display file contents.
10. List all files and directories.
11. Search for report files.
12. Search specific content inside files.
13. Copy report files to Backup directory.
14. Rename files.
15. Move files between directories.
16. Delete a file.
17. Delete an empty directory.

Objectives:

- To understand basic Linux system administration commands.
- To perform file and directory management operations.
- To create, copy, move, rename, and delete files.
- To search files and text using Linux commands.
- To monitor system information and resources.
- To perform backup and basic file management operations.

Observation / Output

```
(tushar@Tushar-laptop)~  
$ whoami  
tushar
```

```
(tushar@Tushar-laptop)~  
$ pwd  
/home/tushar
```

```
(tushar@Tushar-laptop)~  
$ uname -a  
Linux Tushar-laptop 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
```

```
(tushar@Tushar-laptop)~  
$ uptime  
15:12:09 up 51 min, 5 users, load average: 1.10, 0.97, 0.58
```

```
(tushar@Tushar-laptop)~  
$ mkdir MarketingAnalytics
```

```
(tushar@Tushar-laptop)~/MarketingAnalytics  
$ mkdir RawData ProcessedData Reports  
  
(tushar@Tushar-laptop)~/MarketingAnalytics  
$ ls  
ProcessedData RawData Reports
```

```
(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ touch RawData/sales.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ touch RawData/customers.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ touch Reports/report1.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ touch Reports/report2.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ find .
.
./RawData
./RawData/customers.txt
./RawData/sales.txt
./ProcessedData
./Reports
./Reports/report2.txt
./Reports/report1.txt
```

```
(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ echo "Sales Report for July" > RawData/sales.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ echo "Customer Data 2026" > RawData/customers.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ echo "Marketing Analytics Report" > Reports/report1.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ echo "Performance Analysis Completed" > Reports/report2.txt
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ cat RawData/sales.txt
Sales Report for July

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ cat RawData/customers.txt
Customer Data 2026

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ cat RawData/report1.txt
cat: RawData/report1.txt: No such file or directory

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ cat Reports/report1.txt
Marketing Analytics Report

(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ cat Reports/report2.txt
Performance Analysis Completed
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ ls -R
.:
ProcessedData  RawData  Reports

./ProcessedData:

./RawData:
customers.txt  sales.txt

./Reports:
report1.txt    report2.txt
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ find . -name "report*"
./Reports/report2.txt
./Reports/report1.txt
```

```
(tushar@Tushar-laptop)-[~/MarketingAnalytics]
$ grep -r "Report" .
./RawData/sales.txt:Sales Report for July
./Reports/report1.txt:Marketing Analytics Report
```

```
(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ mkdir Backup

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ cp Reports/report1.txt Backup/

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ cp Reports/report2.txt Backup/

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ ls Backup
report1.txt  report2.txt
```

```
(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ mv Reports/report1.txt Reports/FinalReport.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ ls Reports
FinalReport.txt  report2.txt
```

```
(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ mv RawData/customers.txt ProcessedData/

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ ls RawData
sales.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ ls ProcessedData
customers.txt
```

```
(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ rm Reports/report2.txt

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ ls Reports
FinalReport.txt
```

```
(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ mkdir Temp

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ rmdir Temp

(tushar@Tushar-laptop)~[/MarketingAnalytics]
$ ls
Backup  ProcessedData  RawData  Reports
```

Key Questions /Analysis:

1. What is the difference between absolute and relative paths?

Absolute Path: Starts from the root directory (/) and specifies the complete location of a file.

Example: /home/student/MarketingAnalytics/Reports/report.txt

Relative Path: Starts from the current working directory.

Example: Reports/report.txt

2. Why are Linux permissions important for file security?

Linux permissions control who can read, write, or execute files and directories. They protect sensitive information from unauthorized access and improve system security.

3. What is the purpose of the grep command?

The grep command searches for specific words or patterns inside files. It is commonly used to filter logs, configuration files, and text data.

Example: grep "CustomerID" customer_data.txt

4. How does Linux support efficient file management?

Linux provides powerful commands such as mkdir, cp, mv, rm, find, grep, and ls for efficient file organization, searching, backup, and management.

5. Why are backups important in enterprise environments?

Backups protect important business data from accidental deletion, hardware failure, ransomware attacks, and system crashes. They ensure business continuity and quick data recovery.

Supplementary Problems:**1. Create a shell script to automate project directory creation.**

```
(tushar@Tushar-laptop)-[~]  
$ nano create_project.sh  
  
(tushar@Tushar-laptop)-[~]  
$ chmod +x create_project.sh  
  
(tushar@Tushar-laptop)-[~]  
$ ./create_project.sh  
Project directories created successfully.  
  
(tushar@Tushar-laptop)-[~]  
$
```

2. Find the top 10 largest files in the system.

```
(tushar@Tushar-laptop)-[~]  
$ du -ah . | sort -rh | head -10  
1.9G .  
1.2G ./hadoop  
912M ./hadoop/share  
497M ./hadoop/share/hadoop  
492M ./hadoop-3.4.3.tar.gz  
416M ./hadoop/share/doc/hadoop  
416M ./hadoop/share/doc  
311M ./hadoop/lib/native  
311M ./hadoop/lib  
207M ./hadoop/share/doc/hadoop/hadoop-yarn
```

3. Schedule a backup task using cron.

```

(tushar@Tushar-laptop)-[~]
$ crontab -e
no crontab for tushar - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano          ← easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny

Choose 1-3 [1]: 1
crontab: installing new crontab

(tushar@Tushar-laptop)-[~]
$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
0 2 * * * cp -r ~/MarketingAnalytics ~/Backup

```

4. Monitor CPU utilization continuously using top/htop.

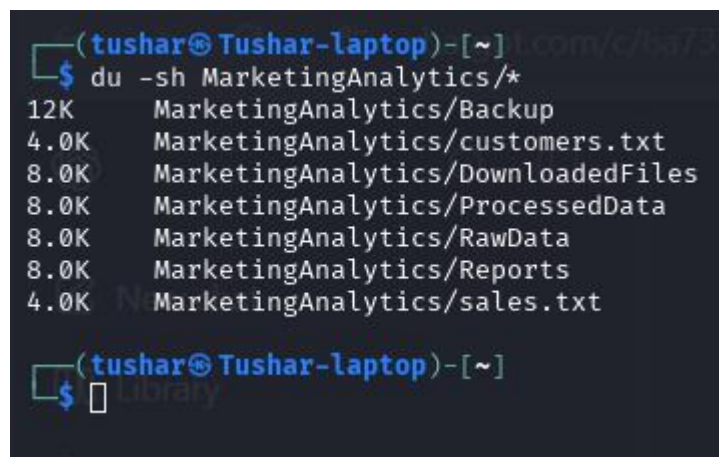
```

(tushar@Tushar-laptop)-[~]
$ top
top - 22:41:27 up 12 min, 1 user, load average: 0.00, 0.17, 0.15
Tasks: 87 total, 1 running, 86 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.2 sy, 0.0 ni, 99.5 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 7783.7 total, 6429.0 free, 1145.4 used, 437.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 6638.3 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM    TIME+  COMMAND
 2520 tushar    20   0 1130608 96056 63992  S   2.0   1.2   0:03.27 qterminal
 564  tushar    20   0 386820 129952 70780  S   0.7   1.6   0:05.70 Xtigervnc
 807  tushar    20   0 277068 30184 22852  S   0.7   0.4   0:02.26 wrapper-2.0
 626  tushar    20   0 2285148 108016 84860  S   0.3   1.4   0:02.58 xfwm4
 805  tushar    20   0 401312 148740 22060  S   0.3   1.9   0:02.13 wrapper-2.0
 941  tushar    20   0 485912 22792 19712  S   0.3   0.3   0:00.06 xfce4-notifyd
    1 root      20   0 25696 15508 11508  S   0.0   0.2   0:00.64 systemd
    2 root      20   0 3180 2212 2072  S   0.0   0.0   0:00.00 init-systemd(ka
    7 root      20   0 3496 2432 2008  S   0.0   0.0   0:00.11 init
   46 root      20   0 50396 16352 14924  S   0.0   0.2   0:00.21 systemd-journal
   63 root      20   0 38948 13752 9352  S   0.0   0.2   0:00.44 systemd-udev
  120 root      20   0 4308 2612 2564  S   0.0   0.0   0:00.00 cron
  122 message+ 20   0 9416 5984 3944  S   0.0   0.1   0:00.36 dbus-daemon
  138 polkitd    20   0 382016 9948 7424  S   0.0   0.1   0:00.20 polkitd
  145 root      20   0 17832 9272 8012  S   0.0   0.1   0:00.00 systemd-logind
  409 root      20   0 308444 7572 6828  S   0.0   0.1   0:00.00 lightdm
  412 root      20   0 5232 2792 2592  S   0.0   0.0   0:00.00 agetty
  421 root      20   0 3196 1120 980  S   0.0   0.0   0:00.00 SessionLeader
  422 root      20   0 3196 1252 1108  S   0.0   0.0   0:00.01 Relay(423)
  423 tushar    20   0 8440 5520 3852  S   0.0   0.1   0:00.03 bash
  424 root      20   0 7628 4440 3880  S   0.0   0.1   0:00.00 login
  434 tushar    20   0 24500 13928 11076  S   0.0   0.2   0:00.20 systemd
  436 tushar    20   0 22596 3940 2156  S   0.0   0.0   0:00.00 (sd-pam)
  474 tushar    20   0 7996 4552 4020  S   0.0   0.1   0:00.02 dbus-daemon
  475 tushar    20   0 94528 8088 6380  S   0.0   0.1   0:00.02 pipewire
  476 tushar    20   0 5336 4872 3332  S   0.0   0.1   0:00.02 bash
  479 tushar    20   0 7256 3712 3472  S   0.0   0.0   0:00.00 mpris-proxy
  480 tushar    20   0 396996 14588 11492  S   0.0   0.2   0:00.17 wireplumber

```

5. Generate a disk usage report.



```
(tushar@Tushar-laptop)-[~]
$ du -sh MarketingAnalytics/*
12K   MarketingAnalytics/Backup
4.0K  MarketingAnalytics/customers.txt
8.0K  MarketingAnalytics/DownloadedFiles
8.0K  MarketingAnalytics/ProcessedData
8.0K  MarketingAnalytics/RawData
8.0K  MarketingAnalytics/Reports
4.0K  MarketingAnalytics/sales.txt

(tushar@Tushar-laptop)-[~]
```

Tools /Technology used:

- Ubuntu Linux / CentOS Linux
- Linux Terminal
- Bash Shell
- Linux File System

Conclusion :

The practical successfully demonstrated Linux system administration and file management operations. Various Linux commands were used to create directories, manage files, search data, monitor system resources, and perform backup operations. This practical improved understanding of Linux command-line operations, file organization, and system administration concepts required in enterprise environments.